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Short-Window Sensitivity of WRF Cold-Pool State to Driving Data during a Uinta Basin Winter Ozone Episode

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Abstract

Cold-pool structure controls the mixing environment available to winter-ozone chemistry, but aggregate surface skill alone does not establish that structure. We evaluated deterministic sensitivity in Weather Research and Forecasting simulations of the 2 February 2013 Uinta Basin episode, anchored by four runs that crossed GFS and NAM driving products with one-way and two-way nest feedback. Across five post-spin-up hours, the GFS members were warmer at the surface, about 3.6 K less stable from 50 to 500 m above ground, and 0.8–1.0 m s^{-1} faster in fixed-layer transport wind than the feedback-matched NAM members. They also had about 1.2 MJ m^{-2} less footprint-mean heat deficit, but every simulation gained heat deficit during the window, identifying a state offset rather than faster GFS cold-pool loss. Nest-feedback effects were much smaller, and a controlled boundary-layer/surface-layer treatment produced a smaller, metric-dependent response. The design does not isolate initial from lateral-boundary forcing, validate inversion depth or event-scale persistence, calculate basin export, or predict ozone.

Keywords: winter ozone; WRF; cold-air pool; Uinta Basin; driving data; nest feedback; heat deficit; model evaluation

1. Introduction

Winter ozone in the Uinta Basin develops when photochemistry proceeds within snow-covered, stably stratified air that restricts vertical dilution and horizontal exchange [1–3]. Snow increases actinic flux and modifies deposition, while the stable, weakly ventilated air confines ozone and its precursors near the basin floor. For a coupled model, the meteorological question is therefore not only whether 2 m temperature has a small mean error. It is whether the atmosphere delivered to the chemistry grid contains a credible shallow mixing environment and weak transport during the accumulation period.

Published concentration agreement does not provide a clean ranking of the relevant uncertainties. Ahmadov et al. obtained a much stronger ozone simulation after applying an aircraft-derived, Horsepool-centered emissions scaling while holding the meteorological simulation fixed, and they conditioned the resulting inventory interpretation on adequate meteorology [1]. Matichuk et al. evaluated a cumulative ladder that changed volatile-organic-compound emissions, leaf area, and deposition together, while improved WRF temperature profiles did not consistently improve ozone profiles [4]. These outcomes are consistent with compensating error among confinement, emissions, photolysis, deposition, and chemical mechanism. They make meteorological evaluation necessary for chemical attribution without making meteorology the uniquely dominant error source.

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The meteorological literature also warns against reducing an experiment to a configuration label. Neemann et al. showed strong case-specific responses from bundled snow, albedo, and cloud-microphysics choices, but their ice-retention modification was a bounding treatment and the 1.33 km grid evaluated most closely was finer than the 4 km meteorology supplied to the chemistry model [2]. Tran et al. showed that the meteorological response to four-dimensional data assimilation depended on the nudged variables, input data, and diagnosed boundary-layer depth; their disclosed artificial formaldehyde limits quantitative chemical inference [5]. Thus, labels such as "NAM driven," "nudged," or "high resolution" do not by themselves identify a controlled cause, and aggregate scores can conceal a failure to retain the layer of interest.

We therefore treat the archived runs as a graph of controlled contrasts rather than a chronological configuration ladder. Our first question is whether the short-window lower-layer state responds reproducibly to the GFS-versus-NAM driving-data product under both nest-feedback modes. Our second is whether the product separation appears in a footprint-integrated heat-deficit measure and represents a state offset or different within-window evolution. Our third is how that repeated response compares with controlled boundary-layer/surface-layer, slope-radiation, feedback, and static-terrain treatments already present in the archive. The analysis evaluates the meteorological state supplied to a later chemistry calculation; it does not simulate chemistry, determine a generally superior product, establish event-scale persistence, or infer basin export.

2. Materials and Methods

2.1. Case, Observations, and Evaluation Sample

The target is the snow-covered Uinta Basin cold-air pool on 2 February 2013 within the episode examined by prior meteorological and chemical studies [1,2]. All model comparisons use 14:00–18:00 UTC, or 07:00–11:00 mountain standard time. The first two forecast hours were excluded as spin-up, leaving five common hourly samples. The sensitivity of the result to that two-hour choice has not yet been evaluated.

Surface observations were extracted from the Synoptic archive for eleven stations reporting 2 m temperature, ten reporting 10 m wind speed, and nine reporting surface pressure during every retained hour. Subhourly reports were averaged within each UTC clock hour. The fixed sample includes basin-floor, sidewall, and bench locations, but it is neither spatially uniform nor area weighted. Every controlled simulation archived both the 1 km d02 and the 333 m d03 during this window, so output availability did not determine the evaluation domain. The d02 was selected because it contains all eleven stations, whereas seven stations clamp to the boundary of the roughly 50 by 48 km d03 and only four provide usable interior columns. The result therefore describes the full-network d02 sample and is not a d02–d03 resolution comparison. Observed temperature from 31 January 00:00 UTC through 4 February 00:00 UTC provides multi-day context around the modeled window.

Observed and modeled surface potential temperature was calculated as $\theta = T(1000/p)^{0.2854}$, with T in kelvin and p in hPa. The elevation-spanning contrast was defined as Mountain Home on the northwest bench minus Pariette Draw on the basin floor. Because the stations are horizontally separated and locally exposed surface sites, this difference is a structural index rather than a colocated free-air inversion strength or depth. No sounding, profiler, or remote-sensing profile is used to validate the modeled 50–500 m layer.

Episode-scale ozone context came from the current EPA Air Quality System (AQS) AirData daily archive for 1 January–28 February 2013. We retained the EPA-reported "Ozone 8-hour 2015" daily-summary row at Ouray, Redwash, Roosevelt, Vernal, and Little Mountain and converted its first maximum value from parts per million to parts per billion

without recomputing an eight-hour statistic. All five sites contained one daily row on each of the 59 local calendar dates; within-day observation coverage ranged from 47% to 100% and is marked explicitly in the figure. The satellite panel uses the NASA Global Imagery Browse Services Terra MODIS corrected-reflectance bands 7-2-1 layer for 2 February 2013. That layer is a calendar-day browse image rather than a quantitative snow retrieval or a crop of the selected 18:15–18:20 UTC granule. These observations establish chemical and snow context only and do not validate the WRF layer or attribute ozone to the simulated meteorology.

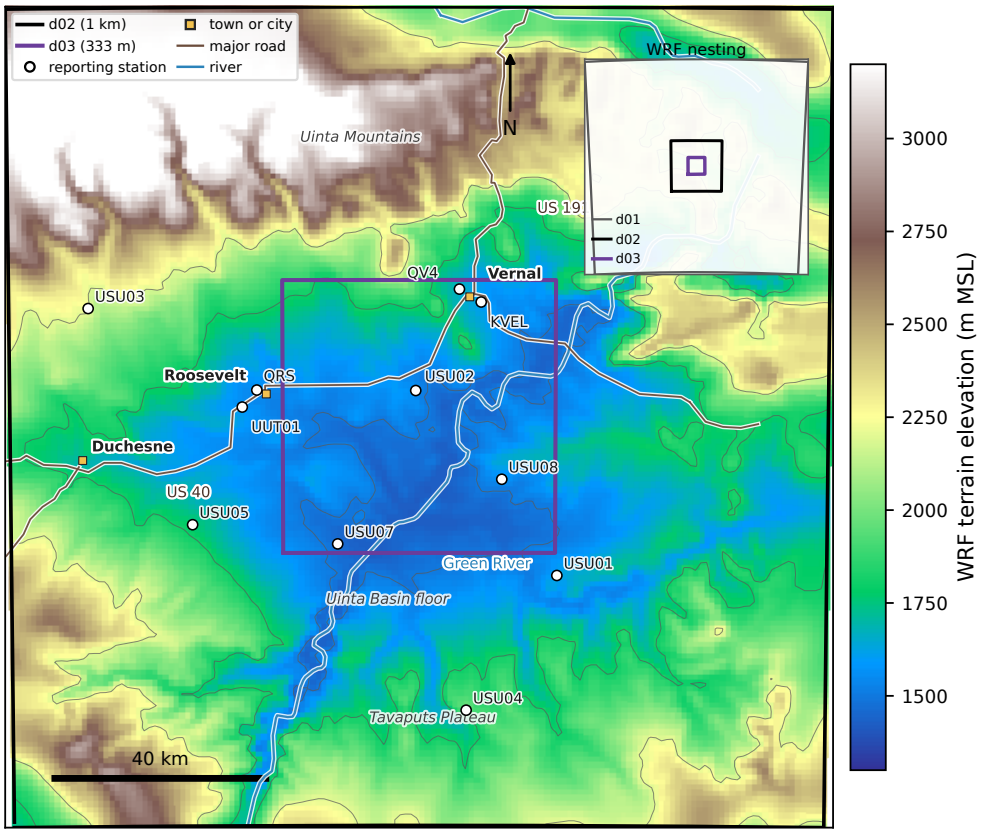


Figure 1. Geographic and sampling orientation for the evaluation. The main map places the d02 terrain, d03 interior, reporting stations, major roads, rivers, towns, and regional labels on one geographic frame. The inset shows the d01–d02–d03 nesting without duplicating the station panel. Seven of the eleven reporting stations fall outside the usable d03 interior, which is why d02 supplies the full-network common sample; the domain outlines do not constitute a resolution evaluation.

[Data-methods completion item: add the curated Synoptic, EPA AQS, NASA GIBS, and MODIS dataset citations; complete Synoptic quality-control parameters, station ownership and sensor heights, observation uncertainty, and reuse terms before coauthor circulation.]

2.2. WRF Experiment Matrix

The four simulations used WRF version 4.8.0 and ran from 12:00 to 18:00 UTC on 2 February 2013. Three nested domains had horizontal grid spacing of 3, 1, and 0.333 km, 75 eta levels, and a 50 hPa model top. The physics suite comprised Thompson microphysics, RRTMG longwave and shortwave radiation, the revised MM5 surface layer, the Noah land-surface model, and the YSU planetary-boundary-layer scheme. All simulations were free forecasts with grid and observation FDDA disabled. They used the same domain geometry,

default 30-arc-second-class terrain field, physics, initialization time, output cadence, and valid window.

The factorial crossed the archived GFS analysis product `gfsanl_4` with the NAM analysis product `namanl_218`, and crossed one-way (`feedback=0`) with two-way (`feedback=1`) nest feedback (Table 1). The products supplied zero-hour analyses at 12:00 and 18:00 UTC and had 27 and 40 metgrid levels for GFS and NAM, respectively. Thus, the driving-data contrast is a product-level treatment that includes the analyses' native horizontal and vertical information; it is not an isolated perturbation of one atmospheric variable. Within each product family, normalized as-run namelists differed only in the feedback flag. The GFS one-way run reused the six verified metgrid files from the GFS two-way run before rerunning `real.exe` and `wrf.exe`.

Table 1. Frozen driving-product by nest-feedback experiment matrix. Stable study codes link each member to the run and comparison inventories.

Study code	Analysis product	Nest feedback	Manuscript label
X6	GFS	two-way	GFS two-way
X10	GFS	one-way	GFS one-way
X5	NAM	two-way	NAM two-way
X3	NAM	one-way	NAM one-way

2.3. Secondary Controlled Contrasts and Design Exclusions

The four anchor simulations are part of a larger reconciled archive in which additional pairs isolate one treatment at a time. The controlled secondary contrasts change the boundary-layer/surface-layer schemes, slope-aware radiation and shading, nest feedback at fine terrain, or static-terrain source (Table 2). Each pair retains its registered differencing direction and its predeclared gate outcome. The archived X7, X0, and X8 configurations cannot form a resolution ladder because they differ simultaneously in initialization and forecast age, domain count and placement, feedback, vertical levels, terrain treatment, radiation options, and output cadence. X7 and X8 therefore remain design or within-run process context and do not enter the comparative Results.

Table 2. Secondary controlled contrasts carried into the response hierarchy. "Failed" preserves the predeclared outcome; "diagnostic" means no criterion was declared before inspection.

Contrast	Operation	Isolated treatment	Evidential role
CMP08	X2 − X1	MYJ/Eta versus YSU/MM5	confirmed station response; no scheme ranking
CMP07	X1 − X0	slope radiation and shading	screen failed on a measurement-limited criterion
CMP06	X9 − X0	two-way versus one-way feedback at fine terrain	confirmed bounded feedback response
CMP09	X3 − X0	default versus fine static terrain	predeclared geometry gate failed
CMP10/CMP11	X3/X0 − X4	default/fine versus coarse static terrain	controlled diagnostic; no predeclared gate

[Model-methods completion item: add the WRF and exact GFS/NAM product citations, map projection, domain dimensions and centers, boundary interval, time step, static-geography versions, and snow-field initialization from the archived as-run records.]

2.4. Surface, Retention, and Ventilation Diagnostics

Surface temperature, earth-relative 10 m wind, and surface pressure were sampled at the nearest d02 cell to each station. Across the 11 cells, model terrain minus station elevation had a median of -2.0 m and a range of -32.4 to +14.3 m. This close elevation match reduces one source of representativeness error but does not account for local exposure or unresolved terrain.

Mass-level height above ground was calculated from perturbation and base-state geopotential and the local WRF terrain height. Grid-relative winds were destaggered and rotated to earth-relative components. Potential temperature and both wind components were then linearly interpolated every 25 m from 50 to 500 m above ground level (AGL) at each station column. Fixed-layer stability was defined as $S_{50:500} = \theta(500\,\mathrm{m}) - \theta(50\,\mathrm{m})$. The transport-wind proxy was $U_{50:500} = \{\bar{u}^2 + \bar{v}^2\}^{1/2}$, where the bars denote height means from 50 to 500 m AGL. This proxy represents the mean wind available for horizontal transport within a fixed lower layer. It is not a calculation of flow across the basin boundary, pollutant residence time, cold-pool volume, or a chemistry-model ventilation coefficient. Native WRF PBL height was retained as a scheme-dependent companion diagnostic rather than interpreted as an observed layer depth.

2.5. Aggregation and Short-Sample Interpretation

Run differences were paired by station and valid hour before spatial aggregation. The driving-product-by-feedback interaction was defined as $[\mathrm{GFS}_{1\mathrm{w}} - \mathrm{GFS}_{2\mathrm{w}}] - [\mathrm{NAM}_{1\mathrm{w}} - \mathrm{NAM}_{2\mathrm{w}}]$ and was likewise evaluated at each station and hour before spatial aggregation. The fixed station field was reduced to one unweighted spatial median per hour, and the five hourly medians were then averaged for each effect summary. The stations were therefore kept together as one spatial field rather than treated as independent replicates. Hourly ranges, sign counts, and leave-one-hour summaries accompany the means.

A two-hour circular moving-block resample of the five hourly medians was also repeated 20,000 times as a descriptive sensitivity calculation. Those ranges are retained in the reproducibility record but are not confidence intervals, hypothesis tests, or estimates of observation, forcing, or structural model uncertainty. No formal population inference is attempted from four deterministic integrations of one case. The fixed two-hour spin-up exclusion has not itself been subjected to a start-time sensitivity test.

2.6. Footprint-Integrated Heat Deficit

Let p_s and p_c denote the pressures at the surface and fixed crest reference, respectively, with $p_c < p_s$. The area-normalized atmospheric heat-deficit proxy was defined as $H = (c_p/g) \int_{p_c}^{p_s} \max\{\theta_{\mathrm{crest}} - \theta(p), 0\}\, dp$, where θ_{crest} is potential temperature interpolated to the crest height in each modeled column. H is reported as a cell-area-weighted mean over an evaluation footprint in MJ m^{-2}. The footprint is the convex hull of the eleven evaluation stations and eleven registered basin waypoints, intersected with model terrain below the crest reference. The primary definition uses a 2200 m crest and the combined hull, covering 4965 d02 cells and 5277 km^2. Six alternate definitions vary the crest to 2100 and 2300 m, dilate or erode the boundary by one cell, or restrict the hull to the stations or waypoints alone. The 2200 m value is the central predeclared reference in that bracket, not an observed cold-pool top or a hydrologic-basin boundary. Because two-way feedback modifies the d02 terrain field by up to 23.1 m relative to the static one-way field, each feedback-matched GFS-minus-NAM pair used a common mask fixed from the one-way NAM terrain. The 13:00 UTC values provide trajectory context only, and all decisions use 14:00–18:00 UTC. A weaker-state criterion required both GFS-minus-NAM pairs to be negative at every retained hour under all seven definitions. A faster-loss criterion required both GFS changes to be

negative and more negative than the feedback-matched NAM changes under all seven definitions. The fixed crest, nonnegative clipping, convex-hull footprint, and modeled profiles make this a bounded proxy rather than an observed cold-pool volume or a direct reproduction of a published basin diagnostic.

3. Results

3.1. Observed Episode Context

The MODIS browse layer shows extensive snow or ice across the basin and surrounding terrain on 2 February, while the AQS series places the modeled day within a sequence of changing daily ozone maxima (Figure 2). On 2 February, the reported daily maximum eight-hour values were 56 ppb at Little Mountain, 65 ppb at Vernal, 72 ppb at Roosevelt, 81 ppb at Redwash, and 95 ppb at Ouray. Each selected monitor reported higher daily values on at least one other date in January or February, so the modeled date is episode context rather than a claim to have sampled the two-month maximum. Neither panel attributes the observed ozone to WRF meteorology.

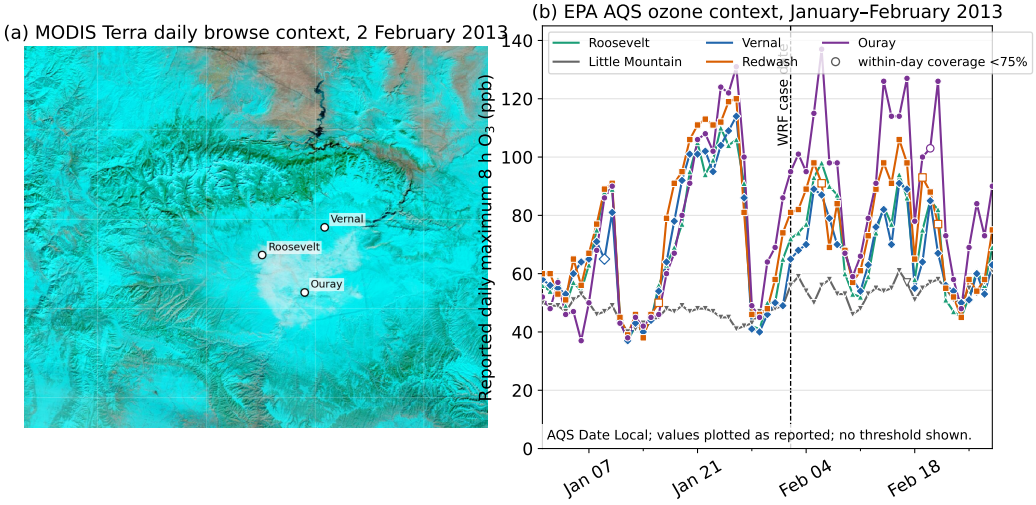


Figure 2. Observed context for the 2 February 2013 simulation date. (a) NASA GIBS Terra MODIS corrected reflectance, bands 7-2-1, for the 2 February calendar day; snow and ice generally appear cyan, most liquid cloud appears white, and ice cloud may also appear cyan. The browse layer is not a raw-granule crop or quantitative snow retrieval; the nearest center-covering Terra granule was acquired at 18:15–18:20 UTC. (b) EPA AQS first maximum value from the "Ozone 8-hour 2015" daily-summary row at Ouray, Redwash, Roosevelt, Vernal, and Little Mountain, converted from ppm to ppb and plotted by AQS local date without recomputation. All five sites have 59 daily rows; hollow markers identify the seven site-days with less than 75% within-day observation coverage. No regulatory threshold is shown, and the panel is episode context rather than chemistry evaluation or meteorology–ozone attribution.

3.2. Observed Surface-Temperature Context

Pariette Draw on the basin floor repeatedly cooled below the higher sites, while the magnitude and ordering of the station temperatures changed through the diurnal cycle (Figure 3). This recurring spatial separation is consistent with cold-pool conditions during the modeled episode. It does not by itself establish the depth, volume, or vertical temperature structure of the cold pool.

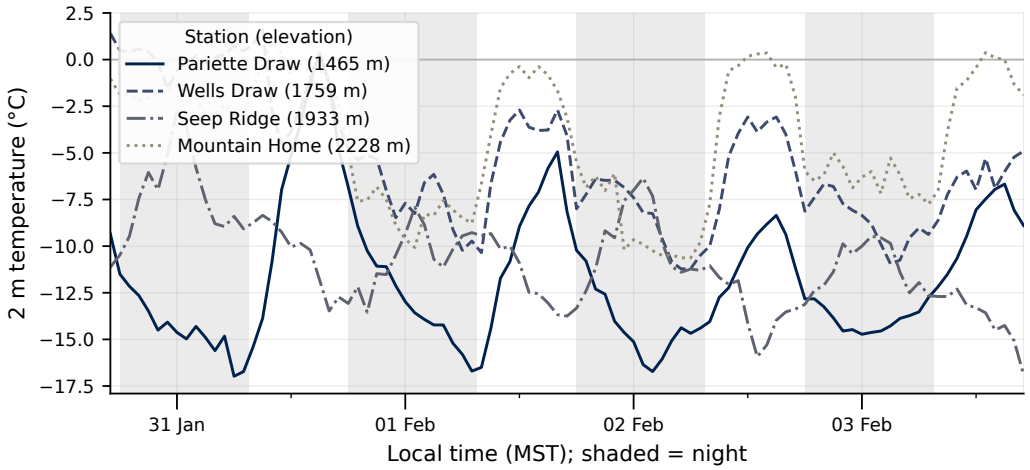


Figure 3. Observed 2 m-temperature evolution at four elevation-spanning Uinta Basin stations from 31 January through 3 February 2013. The shaded intervals indicate approximate local night. The sites provide spatial context but do not form a colocated vertical profile or measure the basin volume.

3.3. Controlled Responses on a Common Pipeline

Figure 4 compares every controlled contrast that shares the station and common-AGL pipeline. The driving-product contrasts are farthest from zero for 2 m-temperature bias, fixed-layer stability, and transport wind. The boundary-layer/surface-layer treatment is smaller but detectable, whereas the slope-radiation, nest-feedback, and interaction effects cluster near zero on those scales. Pressure and wind error do not reproduce that ordering, so this is a response hierarchy rather than a ranking of the best configuration.

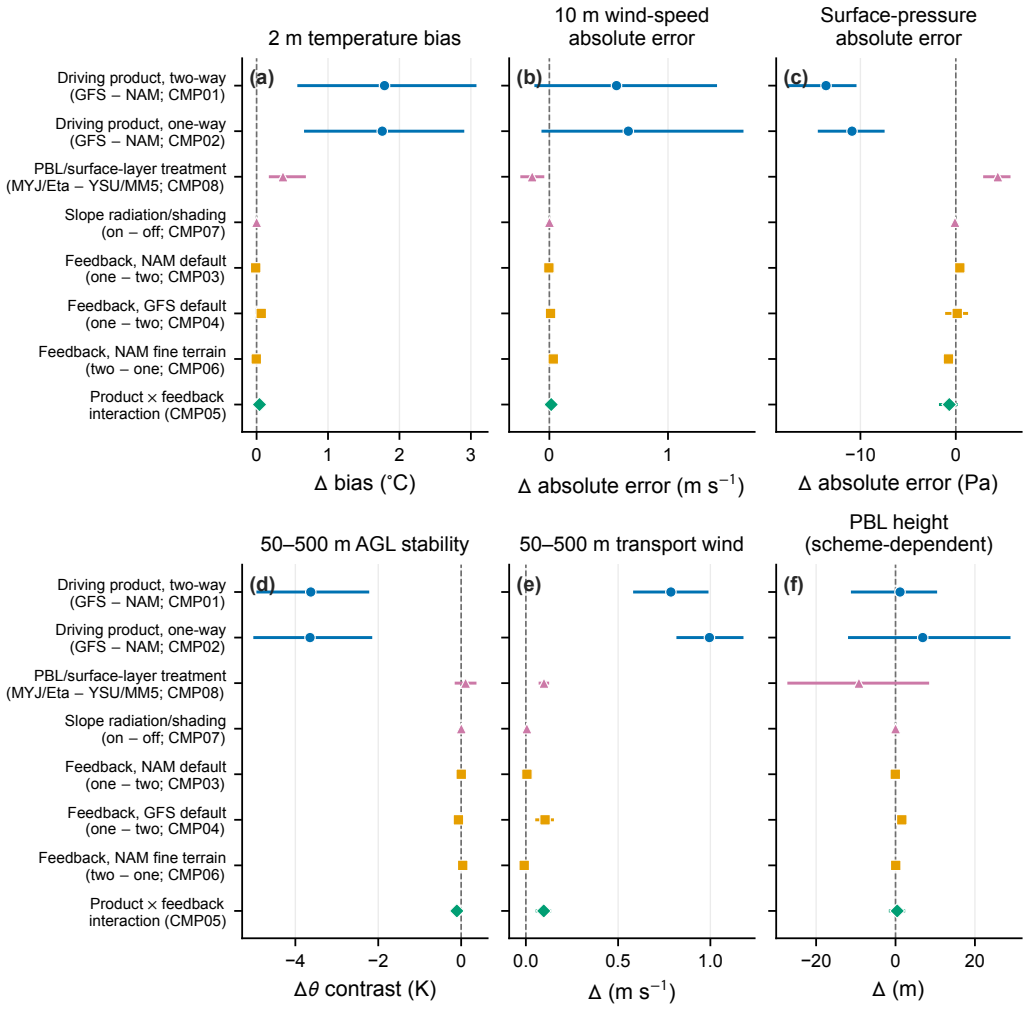


Figure 4. Mean five-hour effects for controlled comparisons that share the station and common-AGL pipeline. Colors and marker shapes identify driving-product, process-treatment, feedback, and interaction contrasts; each row states its differencing direction. Whiskers are descriptive two-hour block-sensitivity intervals for five dependent hourly medians, not confidence intervals. The different ordering of pressure and wind error prevents interpreting the panel as an overall configuration ranking.

3.4. Response to the Driving-Data Product

Under two-way feedback, the GFS-driven run had a smaller Mountain Home-minus-Pariette Draw potential-temperature contrast than the matched NAM-driven run in every retained hour. The observed spatial contrast ranged from 12.63 to 20.21 K, the NAM run from 16.44 to 18.07 K, and the GFS run from 7.35 to 11.74 K. These two-site values are not interpreted as free-air inversion depths.

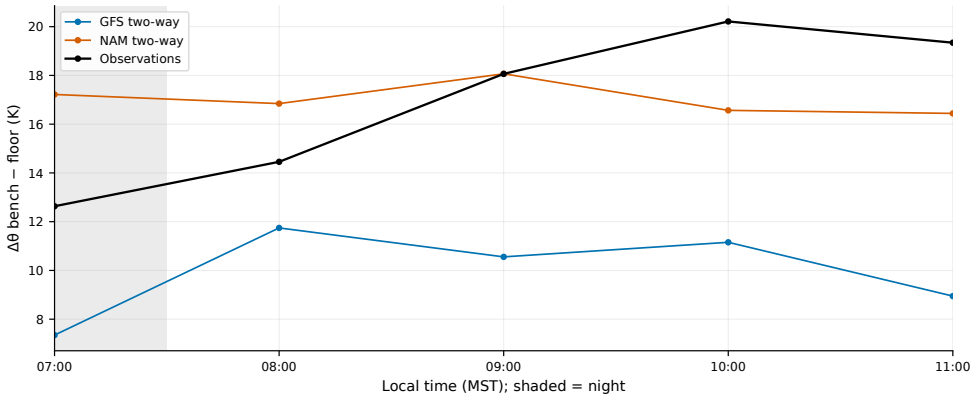


Figure 5. Observed and modeled Mountain Home-minus-Pariette Draw surface potential-temperature contrast from 14:00 to 18:00 UTC on 2 February 2013. The modeled series are GFS two-way and NAM two-way sampled on d02, and all three series contain the same five retained hourly values. The quantity is a horizontally separated bench-minus-floor surface contrast, not inversion depth, a colocated vertical stability measure, or a cold-pool classification threshold.

The five-hour means were 16.94 K for the observations, 17.03 K for NAM two-way, and 9.95 K for GFS two-way. The close NAM value is corroboration for this one surface index, not validation of modeled vertical structure or an overall surface-skill ranking.

The surface evaluation showed the same product ordering under both feedback settings. The mean hourly station-median 2 m-temperature biases were +2.09 and +2.51 °C for the two GFS members, compared with +0.40 and +0.60 °C for the corresponding NAM members. After pairing by station and hour, GFS minus NAM increased the temperature-bias contrast by +1.79 °C under two-way feedback and +1.76 °C under one-way feedback (Table 3). The paired absolute-temperature-error differences were +1.41 and +1.50 °C, respectively. GFS minus NAM also increased 10 m-wind bias by +0.91 and +0.96 m s^{-1}, although the hourly difference in absolute wind error changed sign.

The common-AGL profiles separated the products more directly (Figure 6). Mean GFS-minus-NAM 50–500 m stability was -3.63 K under two-way feedback and -3.65 K under one-way feedback. The corresponding run-level five-hour means were 7.50 and 7.38 K for GFS two-way and one-way, compared with 10.82 K for both NAM members. The respective hourly spatial-median differences ranged from -5.58 to -1.13 K and from -5.36 to -0.90 K, retaining the same sign in all five hours. Mean GFS-minus-NAM transport-wind differences were +0.79 and +0.99 m s^{-1}. The run-level means were 2.01 and 2.06 m s^{-1} for the GFS members, compared with 1.12 and 1.08 m s^{-1} for the NAM members. Their hourly values ranged from +0.37 to +1.18 m s^{-1} and from +0.61 to +1.50 m s^{-1}, again with the same sign in all five hours. In contrast, GFS-minus-NAM PBL-height differences changed sign within both feedback settings.

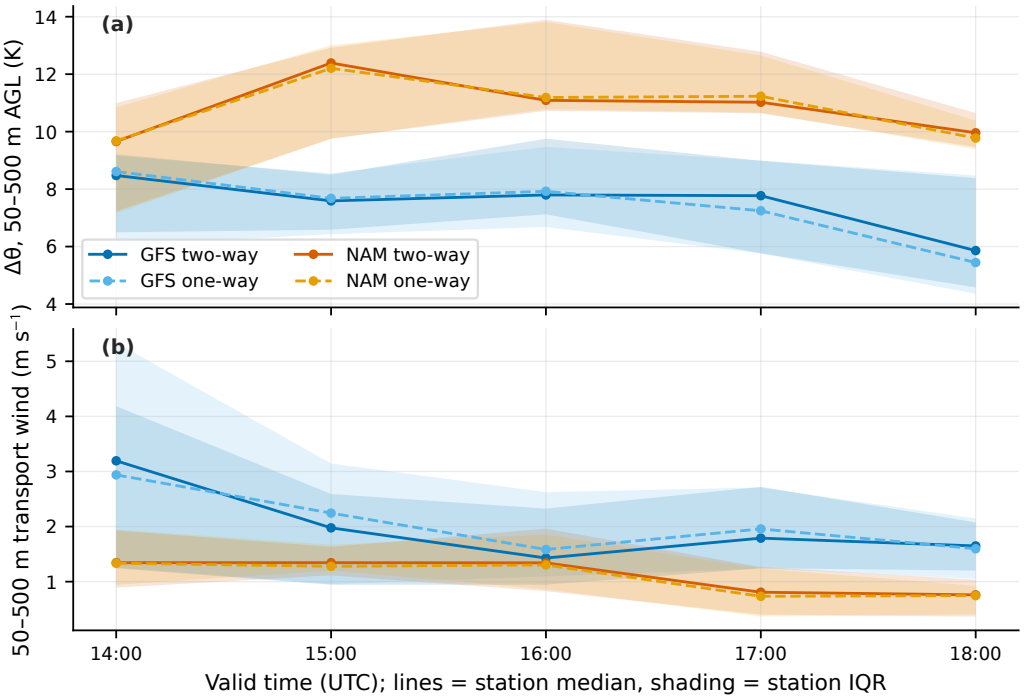


Figure 6. Time evolution of station-column medians and interquartile ranges for 50–500 m AGL potential-temperature stability and fixed-layer transport wind. Solid lines denote two-way feedback and dashed lines denote one-way feedback. The lower-layer GFS–NAM separation recurs under both feedback modes, whereas the within-product feedback response is much smaller.

Table 3. Paired factorial effects averaged over the five hourly spatial medians. Positive temperature values indicate a warmer model-minus-observation bias, positive stability values indicate a more stable fixed layer, and positive transport-wind values indicate a larger layer-mean wind. The effects are deterministic descriptive contrasts rather than inferential estimates.

Effect	ΔT2 bias (°C)	$\Delta S_{50:500}$ (K)	$\Delta U_{50:500}$ (m s^{-1})
GFS − NAM, two-way	+1.79	-3.63	+0.79
GFS − NAM, one-way	+1.76	-3.65	+0.99
GFS one-way − two-way	+0.07	-0.07	+0.10
NAM one-way − two-way	-0.01	+0.00	+0.01
Feedback interaction	+0.04	-0.10	+0.10

3.5. Footprint Heat Deficit: A Weaker State without Faster Loss

The GFS members carried less footprint-mean heat deficit at every retained hour under every tested definition (Figure 7; Table A1). The mean GFS-minus-NAM differences were -1.23 and -1.25 MJ m^{-2} under two-way and one-way feedback. Under the primary definition, five-hour means were 3.00 and 2.97 MJ m^{-2} for GFS two-way and one-way, compared with 4.23 and 4.22 MJ m^{-2} for NAM two-way and one-way. Heat deficit increased in all four simulations, however, so the predeclared faster-loss criterion failed under every definition. The product separation is therefore a state offset already present at 14:00 UTC, not evidence of faster GFS cold-pool loss during the five-hour window.

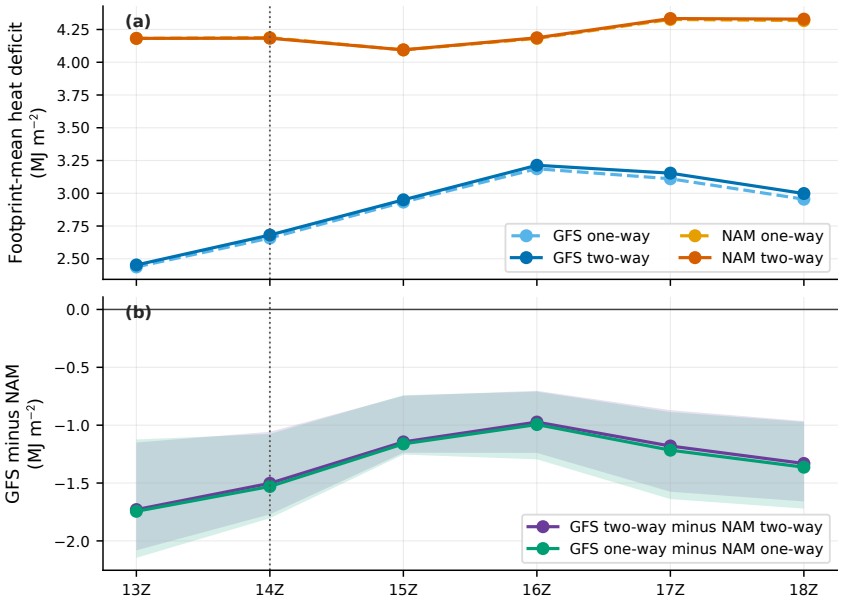


Figure 7. Footprint-mean heat-deficit trajectories for the four default-terrain matrix members and seven-definition sensitivity envelopes for the two GFS-minus-NAM pairs. The 13:00 UTC values are context only; all criteria use 14:00–18:00 UTC. GFS retains a smaller deficit throughout, but the deficit does not decline faster in either GFS member.

3.6. Response to Nest Feedback

The within-product feedback responses were much smaller than the GFS-minus-NAM contrasts, but they were not identical between products (Table 3). For NAM, one-way minus two-way averaged +0.002 K for stability and +0.006 m s^{-1} for transport wind. The NAM hourly effects changed sign and remained within -0.042 to +0.088 K and -0.035 to +0.033 m s^{-1}.

For GFS, one-way minus two-way averaged -0.066 K for stability and +0.104 m s^{-1} for transport wind. The stability effect ranged from -0.198 to +0.067 K and was negative in four of five hours, while the transport-wind effect ranged from +0.025 to +0.197 m s^{-1} and was positive in all five hours. The paired feedback interaction averaged -0.103 K for stability and +0.097 m s^{-1} for transport wind. The interaction stability changed sign once, while its transport-wind component remained positive in all five hours. The 2 m-temperature-bias interaction was +0.041 °C, and the PBL-height interaction also changed sign.

The evidence therefore bounds a small, product-conditioned feedback response on the present d02 sample. It does not establish that one-way and two-way nesting are generally equivalent.

3.7. Secondary Process and Terrain Treatments

The boundary-layer/surface-layer pair changed only MYJ/Eta versus YSU/MM5 while holding fine terrain and slope-aware radiation fixed. MYJ/Eta increased 2 m-temperature bias by 0.37 °C, reduced 10 m-wind absolute error by 0.145 m s^{-1}, and increased fixed-layer transport wind by 0.098 m s^{-1}, with the same sign in all five hours for those metrics (Table A2). These changes were smaller than the driving-product response on the shared lower-layer metrics, and the opposing temperature- and wind-error changes do not support a scheme ranking.

The slope-radiation treatment produced essentially no change at the basin-floor station columns. An aspect-organized skin-temperature response appeared on the sidewalls, but the archived flat-plane shortwave field and missing surface heat fluxes could not test the

mechanism directly. The predeclared screen therefore failed for a measurement-limited reason and supports no improvement or degradation claim.

The fine-terrain feedback pair reproduced the small feedback bound: stability changed by +0.04 K and transport wind by -0.01 m s^{-1}, while both descriptive sensitivity intervals spanned zero (Table A5). The default-minus-fine static-terrain contrast was small on d02 and failed its predeclared terrain-correlation criterion early in the window. The controlled coarse-terrain brackets produced larger diagnostic responses, but no criterion was declared before inspection, so they do not support a terrain-source ranking, drainage pathway, or cold-pool-volume claim.

4. Discussion

4.1. The Main Result Is a Difference in the Supplied Lower-Layer State

The clearest result is not that one simulation wins a surface-score contest. It is that the GFS-driven members supplied a warmer, less stable, and faster-moving 50–500 m layer than the feedback-matched NAM members. The footprint heat-deficit calculation supports the same ordering beyond the fixed station columns. Replication across feedback makes the product separation robust to that configuration choice within this matrix, while the opposite pressure-error ordering prevents a universal product ranking.

This distinction matters to an air-quality model because meteorology sets the layer in which ozone and its precursors mix and move. The GFS state is more consistent with dilution than the matched NAM state, but the transport-wind proxy is not basin export and the heat-deficit footprint is not an observed mixing volume. No chemical species were simulated, so the result identifies a meteorological precondition for attribution rather than an ozone response. That boundary is necessary because emissions, snow and cloud treatment, deposition, chemical mechanism, and meteorological confinement can compensate for one another in coupled simulations [1,2,4].

The driving-product treatment bundles each analysis system's native grid, vertical information, assimilated observations, initial state, and lateral-boundary evolution. The model-only Horsepool profiles at 12:00 and 13:00 UTC expose the supplied states (Figure A1) but neither separate early adjustment from boundary evolution nor validate inversion depth. "Driving-product response" is therefore warranted; "initial-condition effect" is not.

4.2. The Difference Is a State Offset, Not Demonstrated Cold-Pool Loss

GFS had less footprint-mean heat deficit than NAM under every feedback and footprint definition, but deficit increased in every simulation during the retained window. The predeclared faster-loss test therefore failed, and the product separation must be described as a state offset already present by 14:00 UTC. A longer integration through at least one complete diurnal cycle is required to test whether the products differ in cold-pool erosion or persistence.

The registered spatial analysis gives a similarly bounded result. The product response was strongest below 500 m AGL and had the expected footprint-mean sign at every hour, but the two-thirds sign-coherence criterion failed at 14:00–15:00 UTC. The response is shallow and mean-coherent on d02, not spatially uniform throughout the window. Preserving that failed criterion is more informative than relaxing it after inspection.

4.3. Secondary Experiments Narrow Rather than Expand the Claim

The common-pipeline overview makes the response hierarchy explicit on shared metrics. The boundary-layer/surface-layer treatment produced a coherent but smaller response, with temperature error worsening while wind error improved. It is a process-

treatment contrast, not evidence that one named scheme is generally superior. Nest-feedback effects remained small at both default and fine terrain, so feedback protects the product result as a robustness check without becoming a second thesis.

The slope-radiation and fine-terrain screens did not earn figures in the audited manuscript. Slope radiation was nearly invisible at basin-floor stations, while its side-wall skin-temperature signal could not be tested against a slope-corrected radiative or surface-flux diagnostic. The default-minus-fine terrain response was small and failed its predeclared correlation criterion; the wider fine-minus-coarse bracket is controlled and visually stronger, but it had no predeclared scientific gate. The result therefore supports no radiation or terrain-source superiority claim. The multiply confounded apparent resolution ladder is excluded for the same reason: nominal grid spacing cannot be isolated from its other configuration changes.

4.4. Observation and Design Limits Set the Next Experiment

The common-AGL result leads because it applies one geometric layer, station sample, time window, and aggregation to all four anchor simulations. Native PBL height cannot replace it because PBL height is scheme dependent and its product contrasts changed sign. The observed floor-to-bench temperature contrast is useful surface corroboration, but horizontally separated stations do not validate inversion depth, layer stability, or modeled heat deficit. The admitted ozone and satellite panels strengthen episode context without supplying that missing vertical validation or establishing a meteorology–ozone causal response.

The d02 choice is a sampling decision rather than a resolution result. It retains all eleven stations, whereas d03 retains only four usable interior columns, but it may smooth slope, tributary, and terrain-edge responses. A predeclared transfer test must compare d02 and d03 over the same geographic support and station sample before the present hierarchy is assumed to hold at 333 m.

The five-hour window, fixed two-hour spin-up exclusion, dependent hourly summaries, and incomplete Synoptic metadata limit every conclusion. The next discriminating experiment should combine a longer matched product/feedback matrix with an independent in-basin vertical reference and a design that separates initial from lateral-boundary forcing. Only then should selected meteorological states be crossed with emissions and chemical mechanisms to test an ozone response. All failed gates in the present archive remain failed until new evidence is generated under a newly declared criterion.

5. Conclusions

The largest repeated response in the controlled archive is the GFS-versus-NAM driving-product contrast on the common d02 evaluation sample. Relative to NAM, GFS is warmer at the evaluated surface sites, about 3.6 K less stable from 50 to 500 m AGL, 0.8–1.0 m s^{-1} faster in fixed-layer transport wind, and about 1.2 MJ m^{-2} lower in footprint-mean heat deficit under both feedback modes. The heat-deficit separation is a state offset, not demonstrated faster loss, and the surface metrics do not support a universal product ranking. Nest-feedback effects are much smaller, the boundary-layer/surface-layer response is smaller and metric dependent, and the slope-radiation and fine-terrain screens do not pass their declared gates. The present result does not identify the responsible product component, validate inversion depth, establish event-scale persistence, calculate basin export, or predict ozone. Within those bounds, the driving-product treatment produced the largest repeated separation in the lower-basin meteorological state supplied to downstream chemistry during the five-hour window.

Appendix A. Evidence Tables and Supplied-State Profiles

This appendix retains numerical detail needed to audit the concise Results narrative without admitting failed-gate or review-only figures.

Table A1. Footprint-mean heat-deficit contrasts over the 14:00–18:00 UTC decision window. The primary definition uses the combined station-and-waypoint hull with a 2200 m crest; envelopes span all seven registered definitions. Values are deterministic descriptive contrasts in MJ m^{-2}.

GFS − NAM quantity	Two-way pair	One-way pair
Mean difference, primary definition	-1.23	-1.25
Hourly range, primary definition	-1.50 to -0.97	-1.53 to -0.99
Hourly envelope, seven definitions	-1.77 to -0.70	-1.80 to -0.71
GFS change, 14:00–18:00 UTC	+0.32	+0.30
NAM change, 14:00–18:00 UTC	+0.15	+0.13
Difference in change, primary definition	+0.17	+0.17
Weaker-state definitions passed	7/7	7/7
Faster-loss definitions passed	0/7	0/7

Table A2. Controlled process-treatment effects on the frozen station/common-AGL pipeline. Values are means of five hourly spatial medians; signs are positive/negative hours with exact zeros omitted. PBL height is scheme defined across the boundary-layer/surface-layer treatment.

Metric	Slope radiation (X1 − X0)		MYJ/Eta (X2 − X1)	
	Mean	Signs	Mean	Signs
T2 bias (°C)	+0.001	3/1	+0.37	5/0
10 m wind absolute error (m s^{-1})	+0.001	2/2	-0.145	0/5
Surface-pressure bias (Pa)	-0.12	0/4	+6.3	5/0
Stability $S_{50:500}$ (K)	+0.004	3/1	+0.11	3/2
Transport wind $U_{50:500}$ (m s^{-1})	+0.006	4/0	+0.098	5/0
PBL height (m)	+0.01	3/1	-9.2	2/3

Table A3. Surface corroboration for the driving-product contrasts. Every value uses the matched station and hour samples described in Section 2.

Evidence	Result	Interpretation limit
2 m temperature	GFS-minus-NAM bias contrast +1.79 °C two-way and +1.76 °C one-way; absolute-error differences +1.41 and +1.50 °C	Same ordering as common-AGL stability
10 m wind	Bias contrast +0.91 and +0.96 m s^{-1}; hourly absolute-error differences change sign	No stable wind-skill ordering
Surface pressure	Absolute-error contrast -13.6 Pa two-way and -10.9 Pa one-way; negative in every hour	Opposite ordering from temperature, with small absolute differences
Two-site surface index	Observed 12.63–20.21 K; NAM 16.44–18.07 K; GFS 7.35–11.74 K under two-way feedback	Horizontally separated surface contrast, not inversion strength or depth

Table A4. Registered spatial and vertical checks for the two GFS-minus-NAM contrasts over the d02 footprint.

Quantity	Committed range	Criterion and outcome
Mean 2 m potential-temperature difference	+0.89 to +1.16 K at 14:00–15:00 UTC; +2.00 to +2.74 K at 16:00–18:00 UTC	Positive at all ten pair-hours
Footprint sign share	0.64–0.67 at 14:00–15:00 UTC; 0.84–0.93 later	Two-thirds criterion failed at the first two hours
Low-to-aloft absolute-response ratio	1.45–3.12	Greater than one at all ten pair-hours
50–500 m stability difference	-4.33 to -3.04 K	Negative at all ten pair-hours
Gate conjunction	Two of three component criteria passed throughout	Failed because early floor sign coherence missed its threshold

Table A5. Fine-terrain feedback contrast X9 minus X0, or two-way minus one-way. Values are means of five hourly spatial medians; sensitivity intervals are descriptive rather than inferential.

Metric	Mean	Hourly range	Signs (+/−)	Sensitivity interval
Stability $S_{50:500}$ (K)	+0.04	-0.09 to +0.14	4/1	-0.03 to +0.10
Transport wind $U_{50:500}$ (m s^{-1})	-0.01	-0.07 to +0.03	3/2	-0.03 to +0.01
PBL height (m)	+0.05	-0.67 to +0.78	3/2	-0.38 to +0.47
T2 bias (°C)	-0.004	-0.09 to +0.04	3/2	-0.05 to +0.03
10 m wind absolute error (m s^{-1})	+0.03	+0.001 to +0.06	5/0	+0.01 to +0.06
Surface-pressure bias (Pa)	-0.77	-1.23 to -0.35	0/5	-1.06 to -0.50

Table A6. Static-terrain diagnostics retained after the failed default-minus-fine gate and the closed coarse-member lineage check.

Contrast	Terrain difference	2 m response	Status
Default minus fine	7.9 m RMS; -55 to +60 m range	Elevation-band means 0.07–0.27 K; terrain correlation below 0.3 at 14:00–16:00 UTC	Predeclared conjunction failed
Coarse brackets	91.9/93.1 m RMS; extrema to -462/+487 m	Slope-band means 1.2–1.5 K; terrain correlation 0.50–0.67	Descriptive only; no predeclared gate
Coarse-member lineage	As-run namelist matches the controlled family; terrain is materially smoother	Configuration caveat closed	Supports controlled-bracket status, not a terrain ranking

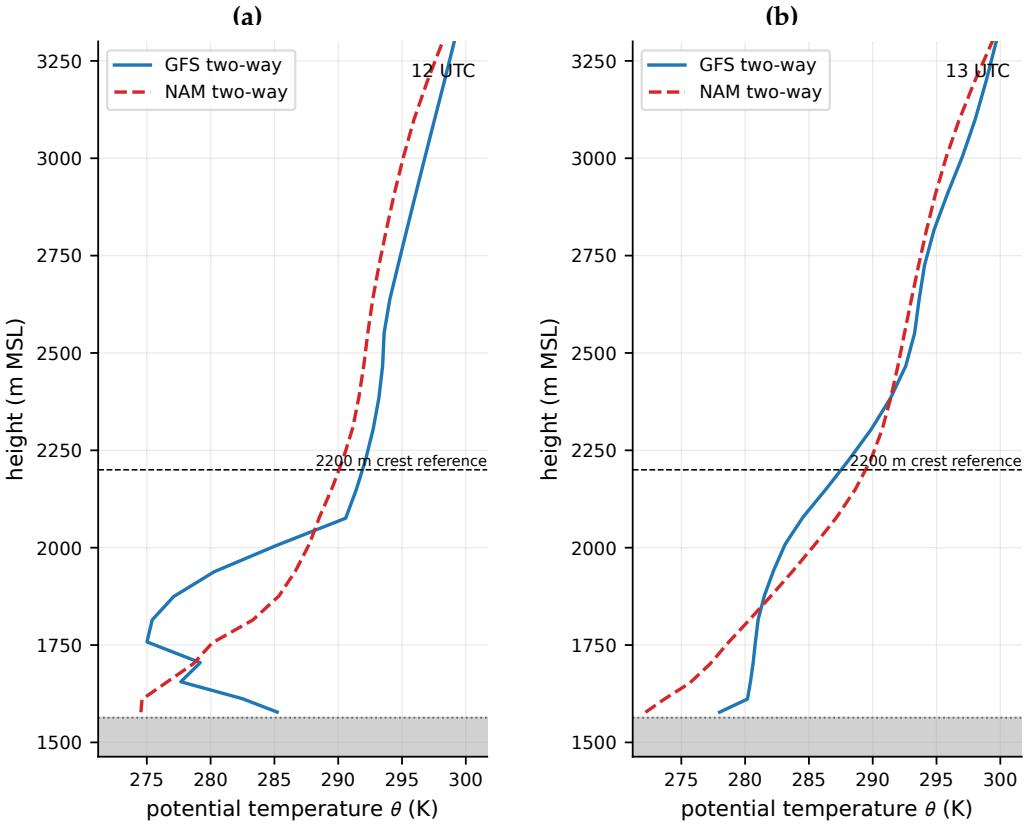


Figure A1. Model-only d03 potential-temperature profiles at Horsepool for GFS two-way and NAM two-way. (a) The 12:00 UTC supplied states and (b) the 13:00 UTC one-hour forecasts use identical axes. Gray shading marks levels below local model terrain, and the dashed 2200 m line is the fixed heat-deficit reference rather than an observed cold-pool top. Changes between panels include both early model adjustment and one hour of lateral-boundary evolution; no colocated observed profile validates the modeled layer.

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